**Department of Electronics and Communication Engineering Faculty Members**

*Birla Institute of Technology, Mesra, Jaipur Campus*

# Dr. Ankur Agarwal

**Joined Institute on: 24-Feb-2015**

**Assistant Professor**

**Electronics and Communication Engg**

**Ph.D. (MNIT, Jaipur), M.Tech (MNIT, Jaipur), B.E. (Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal).**

## CONTACT ADDRESS

27, Birla Institute of Technology, Malviya Industrial Area, Jaipur, Rajasthan 302010

108-B, Umang, Mahanagar, Bareilly, U.P.

Phone Office: 0141 4019827

Phone Residence: 9410890070

Email: ankur@bitmesra.ac.in

## WORK EXPERIENCE

Teaching: 19 Years

## PROFESSIONAL BACKGROUND

EDUCATION QUALIFICATIONS

1. Degree: X

Subject: Science stream

University/Board: U.P. Board

Division/CGPA: Ist

Passing Year: 1994

2. Degree: XII

Subject: Science Stream

University/Board: U.P. Board

Division/CGPA: Ist

Passing Year: 1996

3. Degree: B.E.

Subject: Electronics and Telecommunication Engineering

University/Board: RGPV, Bhopal

Division/CGPA: Ist

Passing Year: 2001

4. Degree: M.Tech

Subject: Electronics and Communication Engineering

University/Board: MNIT, Jaipur

Division/CGPA: Ist

Passing Year: 2009

5. Degree: PhD

Subject: Electronics and Communication Engineering

University/Board: MNIT, Jaipur

Division/CGPA: (Not specified)

Passing Year: 2026

UG COURSES CURRENT SEMESTER

1. Analog Communication

2. Computer Networks

3. Analog Circuits Lab.

PG COURSES

1. Image Processing

2. Image Processing Lab

3. "Digital Image Processing" course for Sequential M.Tech session 2014, at Uttarakhand Technical University, Dehradun, Uttarakhand.

OFFICIAL RESPONSIBILITIES

1. E-Cell Coordinator

EXPERT LECTURE

Delivered expert lecture on "Image Restoration" in one week TEQIP II sponsored workshop entitled "Recent trends in Digital Image Processing & Computer Vision" organised by Department of computer science & engineering, Rajasthan Technical University, Kota, during 18-22, January, 2017.

CERTIFICATION

Cisco Certified Network Associate (CCNA).

SAMSUNG PRISM INDUSTRIAL RESEARCH

Worklet Code: IOTP11

Worklet: AI, ML, Edge computing | Edge AI framework for SmartHome

Name of Faculty Member: Mr. Ankur Agarwal

Name of Coding Qualifying Students: Namrata Chaudhari, BIT, Mesra (Branch IT)

Duration: (AY 2020-22)

Status: Successfully Completed

## RESEARCH AREAS

Integral Transforms (Fractional)

Signal and Image Processing

Wireless Communication

## AWARD AND HONOURS

1. State Board (U.P. Board) scholarship in Xth.

2. State Board (U.P. Board) scholarship in XIIth.

## PUBLICATIONS

1. A. Agarwal and K. K. Sharma, "Medical images fusion using signal decomposition methods and implementation on an embedded platform," in Proc. 5th International Conference on Innovations in Computational Intelligence and Computer Vision (ICICV-2025), CNR NANOTEC, University of Calabria, Italy, June 04–06, 2025, Lecture Notes in Networks and Systems (LNNS), Springer (in press).

2. A. Agarwal and K. K. Sharma, "Empirical Fourier and Variational Mode Decomposition in EEG Seizure Analysis: A Comparative Study with Raspberry Pi Implementation," in Proc. 16th IEEE International Conference on Computing Communication and Networking Technologies (ICCCNT-2025), Indian Institute of Technology Indore, India, July 6–11, 2025, Scopus Indexed (in press).

3. A. Agarwal and K. K. Sharma, "An image processing algorithm for image fusion via variational mode decomposition and its implementation on embedded platforms," J. Opt., 2025.

4. A. Agarwal, "Fusion of Remote Sensing Images using Signal Decomposition Methods: A Comparative Analysis," Journal of Electrical Systems, vol. 20, no. 9s, pp. 2269–2277, 2024.

5. A. Agarwal and S. Mishra, "Peak to average power ratio reduction in sub-carrier Index modulated OFDM using selective mapping," in Proc. International Conference on Computer, Communications and Electronics, 2017.

6. A. Agarwal, D. Agarwal, N. Sharan, and P. R. M., "PAPR Reduction Using Precoding and Companding Techniques for OFDM Systems," in Proc. 2015 International Conference on Advances in Computer Engineering and Applications (ICACEA), IEEE, 2015, pp. 18–23.

7. A. Agarwal, D. Agarwal, and N. Sharan, "Comparative Analysis of OFDM-MIMO System with Golden codes for various antenna multiplexing and modulation schemes," International Journal of Advanced Engineering Research and Technology (IJAERT), vol. 2, no. 6, pp. 168–170, 2014.

8. A. Agarwal, P. Sharma, and D. Mishra, "Efficient image encryption and decryption using discrete wavelet transform and fractional Fourier transform," in Proc. 5th International Conference on Security of Information and Networks (SIN 2012), ACM, New York, USA, 2012, pp. 53–157.

9. A. Agarwal and K. K. Sharma, "Signal Reconstruction using Nonuniform Samples in Fractional Fourier Domains," in Proc. NCVPRIPG, 2010, pp. 186–191.

## WORKSHOPS/ CONFERENCES/TUTORIALS/ FDP

1. Title: Inculcating Universal Human Values in Technical Education (UHV-I)

Duration: 20 Jan – 24 Jan 2025

Organizer: AICTE

Course Type: Online Course

2. Title: Advanced Optimization Technique using MATLAB

Duration: 17 Feb – 28 Feb 2025

Organizer: MeitY / AICTE/ UGC

Course Type: Online Course

3. Title: Remote Sensing and Digital Image Analysis

Duration: 27 Aug – 20 Sept. 2024

Organizer: ISRO/IIRS Dehradun

Course Type: Online Course

4. Title: AI/ML for Geodata Analysis

Duration: 19 Aug. to 23 Aug, 2024

Organizer: ISRO/IIRS Dehradun

Course Type: Online Course

5. Title: Physical Design and Verification Internship - PD16.

Duration: 13.06.2024 to 18.07.2024

Organizer: Cadence Certified Training Partner, Entuple Technologies

Course Type: Internship Training Program

6. Title: Recent Trends in MEMS for 6G Communications

Duration: 17-18 Aug. 2023

Organizer: CSIR, Pilani, Rajasthan

Course Type: FDP

7. Title: VLSI to System Design: Silicon to End Application Approach.

Duration: 31 July - 04 August 2023

Organizer: AICTE, Arm Education and STMicroelectronics

Course Type: Workshop

8. Title: Wireless Communication and Signal Processing Techniques

Duration: 20-24, May 2019

Organizer: MNIT, Jaipur, TEQIP-III

Course Type: Short Term Course

9. Title: 5G and IoT – The New Internet Perspective

Duration: 8-9, December, 2018

Organizer: LNMIIT, Jaipur

Course Type: National Workshop

10. Title: Scientific Techniques & Computational Approaches

Duration: 24-28, December, 2018

Organizer: MNIT, Jaipur, TEQIP-III

Course Type: Short Term Course

11. Title: Wireless and Mobile Communications

Duration: 3-7, December, 2018

Organizer: MNIT, Meity, Government of India.

Course Type: FDP

12. Title: 5G Communication

Duration: 11-15, June, 2018

Organizer: MNIT, Meity, Government of India.

Course Type: FDP

13. Title: LATEX and Technical Writing

Duration: 24-28, January, 2018

Organizer: MNIT, Jaipur

Course Type: Workshop

14. Title: Quantum Communication

Duration: 24, September, 2016

Organizer: TEQIP, MNIT, Jaipur

Course Type: Workshop

15. Title: Digital Hardware Design & FPGA

Duration: 3-16, September, 2016

Organizer: Electronics & ICT Academy, MNIT, Jaipur

Course Type: Academy Training Programme

16. Title: Adaptive Signal Processing: Theory and Applications

Duration: 26 Feb. to 01 March, 2016

Organizer: Electronics & ICT Academy, MNIT, Jaipur

Course Type: Academy Training Programme

17. Title: Compressive Sensing

Duration: 12-13, September, 2015

Organizer: MNIT, Jaipur

Course Type: Workshop

18. Title: Medical Signals and Image Processing

Duration: 13th June – 17th June 2011

Organizer: Indian Institute of Technology, Roorkee.

Course Type: AICTE sponsored QIP course

19. Title: Digital Signal Processing

Duration: 26th Feb – 02nd March 2011

Organizer: Indian Institute of Technology, Roorkee.

Course Type: AICTE sponsored QIP course

20. Title: Satellite Communications Technology and Its Applications

Duration: 16th – 21st March 2009

Organizer: Indian Institute of Technology Kharagpur.

Course Type: AICTE sponsored QIP short term course

21. Title: ADSP and Applications

Duration: 27th January – 06th February 2009

Organizer: National Institute of Technical Teachers Training and Research (NITTTR).

Course Type: AICTE sponsored winter school

22. Title: Vender-delivered training on MATLAB CAD-Tool & specialized tool boxes

Duration: 25th - 27th March 2008

Organizer: MNIT, Jaipur

Course Type: Training

23. Title: Recent Trends in Image Processing

Duration: 23 October 2010

Organizer: Shobhit University, Meerut

Course Type: Workshop

24. Title: Recent Advancement in Microwave Techniques and Applications. (MICROWARE 2006)

Duration: 06-08, 2006

Organizer: University of Rajasthan

Course Type: National conference

25. Title: Multicore technologies and platforms: A road map for India Academia

Duration: 22nd – 23rd September 2006

Organizer: MNIT, Jaipur

Course Type: Conference

26. Title: Optical and Wireless Communication Systems

Duration: April 11, 2008

Organizer: IETE, Jaipur

Course Type: Tutorials Sessions

27. Title: Recent Advances in Broadband Systems

Duration: April 12-13, 2008.

Organizer: IETE, Jaipur

Course Type: 39th IETE mid Term Symposium

## MEMBER OF PROFESSIONAL BODIES

IEEE

# Deepak Chaturvedi

**Joined Institute on: 24-Sep-2009**

**Assistant Professor**

**Electronics and Communication Engg**

**PhD (Pursuing), M.E, B.E.**

## CONTACT ADDRESS

52/60, Mansarovar, Jaipur, Rajasthan

Phone Office: 01414019803

Phone Residence: 9414795959

Email: deepakchaturvedi@bitmesra.ac.in

## WORK EXPERIENCE

Teaching: 28 Years

Individual: 05 Years

## PROFESSIONAL BACKGROUND

Worked as Assistant Manager (Production) in KRNL Jaipur from July 1989 to July 1994 and supervised production, testing of C-DOT based EPABX / EPAX system and PC motherboard.

## RESEARCH AREAS

Electromagnetic Field Effect Measurement.

Microwave Communication Passive Components Design.

Optical Communication System.

Flexible Sensors for next generation EV Battery protection.

## PUBLICATIONS

1. G. Singh, D. Chaturvedi, D. Verma and P. S. Rathore, "Recent Advances in Battery Management System Topologies for Electric Vehicles," 2025 12th National Power Electronics Conference (NPEC), Kozhikode, India, 2025, pp. 1-6, doi: 10.1109/NPEC66512.2025.11450232.

2. S.M.M Quadri, P. Goyal, S. Bhargav, D Chaturvedi; Increasing Front To Back Ratio Of Phased Array Antenna Without Increase In Beam Width For The Mobile Base Station: NCMDC 2012

## MEMBER OF PROFESSIONAL BODIES

Life Member of Broadcast Engineering Society (India) Society from 2007

## TEXT AND REFERENCE BOOKS

Fundamentals of Microwave Engineering: Paragon International Publishers New Delhi

## INTELLECTUAL PROPERTY RIGHTS (IPR)

FLEXIBLE SENSOR SYSTEM FOR PROTECTING BATTERY DEFORMATION AND RELATED METHODS

# Gaurav Jain

**Joined Institute on: 13-Oct-2010**

**Assistant Professor**

**Electronics and Communication Engg**

**Pursuing PhD from MNIT, M.E. (MBM Engineering College 2009), B.E. (Electronics & Communication Engineering) (MBM Engineering College 2001)**

## CONTACT ADDRESS

F-6, D-175, Moti Marg, Bapu Nagar, Jaipur

Phone Office: 01414019803

Phone Residence: 9413256269

Email: gauravjain@bitmesra.ac.in

## WORK EXPERIENCE

Teaching: 17 Years

Research: 2 Years

Individual: 3 Years

## PROFESSIONAL BACKGROUND

EDUCATIONAL BACKGROUND

1. B.E. (Electronics & Communication Engineering) MBM Engineering College, Jodhpur (2001)

2. M.E. (Digital Communication Engineering) MBM Engineering College, Jodhpur 2009.

3. PhD Pursuing MNIT, Jaipur (Raj.)

Have one year experience on DBMS based industry project.

SUBJECTS TAUGHT

Basic Electronics, Analog Communication, Digital Communication, Electronic Instrument and Measurement, Signal and System, Digital Signal Processing

INSTITUTE RESPONSIBILITIES

1. Annual TechVibe Coordinator

2. GATE Advisory Cell Coordinator

3. Mentor of four Students

## RESEARCH AREAS

Research Area: Bio-Medical Instrument, Wireless Communication, Digital Communication

Techniques: Matlab, Micro controller, Embedded C

## PUBLICATIONS

1. Paper presented at IEEE-iSES 2021 at MNIT, Jaipur

G. Jain, A. M. Joshi, R. K. Maddila and S. K. Vipparthi, "A Review of Non-Invasive HbA1c and Blood Glucose Measurement Methods," 2021 IEEE International Symposium on Smart Electronic Systems (iSES), Jaipur, India, 2021, pp. 339-342, doi: 10.1109/iSES52644.2021.00086.

Keywords: {Proteins; Pathology; Red blood cells; Current measurement; Laboratories; Measurement techniques; Glucose; hba1c; glucose; non invasive}

https://ieeexplore.ieee.org/xpl/conhome/9700979/proceeding

2. Paper presented at 5th International Conference on Data Science and Applications at MNIT Jaipur

"Non-Invasive Measurement of HbA1C Using Spectroscopic Method"

July 17-19, 2024, Springer edition

## WORKSHOP/TRAINING/CONFERENCE ATTENDED

1. One week "Academy training Programme on Adaptive signal Processing" by Electronics & ICT Academy held at MNIT campus Jaipur from Feb 26 to March 1, 2016.

2. One week "Short Term Course on Recent Advances in signal Processing and Communication" by TEQIP held at MNIT campus Jaipur from May 19 to 23, 2014.

3. One week "Faculty Development Programme on Advancement in Adaptive signal Processing" by Department of Electronics & Communication Engg held at MNIT campus Jaipur from December 17 to 21, 2013.

4. One week "Short Term Course on Recent Advances in signal Processing and Communication" by Department of Electronics & Communication Engg held at MNIT campus Jaipur from September 1 to 5, 2012.

5. Two week "Training Course on Entrepreneurship Development" by Department of Science & Technology held at Birla Institute Of Technology Jaipur Campus from January 15 to 27, 2013

## CURRENT SPONSORED PROJECTS

PROJECT/THESIS SUPERVISED AT UG LEVEL

1. Controlling Robotic Arm Using Different Modes By Bluetooth Technology (Ms. Aditi Yadav, Bhawana Bakolia, Kajal Kansal), BE(ECE) 2017, BIT Jaipur.

2. Developing a Platform for Face and Expression Recognition Using Matlab (Mr. Shashank Yadav), B.E(ECE) 2017, BIT Jaipur

3. Using IOT Automation of Remote Appliances to Avoid Accidents and Save Energy (Mr. Ragvender Singh, Shivant Shikar, Shashi Bhusan), BE(ECE) 2016, BIT Jaipur

4. Arduino based home automation (Mr. Aman Ashish, Mr. Atul Raj, Mr. Ritesh Kumar, Mr. Mahesh Kumar Jat), BE ECE 2017, BIT Jaipur

# Dr. Jyoti Sharma

**Joined Institute on: 18-Mar-2010**

**Assistant Professor**

**Electronics and Communication Engg**

**PhD. from Malaviya National Institute of Technology (2024), M.Tech. from Malaviya National Institute of Technology (2009), B.E. from University of Rajasthan (2004)**

## CONTACT ADDRESS

48/46, Moti Path, Rajat Path, Mansarovar, Jaipur, Rajasthan

Phone Office: 9929655006

Phone Residence: 9929655006

Email: jyotisharma@bitmesra.ac.in

## WORK EXPERIENCE

Teaching: 21 Years

## PROFESSIONAL BACKGROUND

TEACHING EXPERIENCE

15 years teaching experience at BIT Mesra, Jaipur Campus (2010 onwards)

4 years teaching experience at RCEW, Bhankrota, Jaipur (2006-2010)

2 years teaching experience at RCERT, Jaipur (2004-2006)

SUBJECT TEACHING IN CURRENT SEMESTER

EC261 Digital Computer Electronics (IV AI)

EC305 Digital Signal Processing (IV EEE)

EC306 Digital Signal Processing Lab (IV EEE)

CS338 Embedded Systems Lab (VI CSE)

SUBJECTS TAUGHT IN PREVIOUS SEMESTER

EC319 VLSI Systems (V ECE)

EC101 Basic Electronics (I Sem)

EC202 Electronic Devices Lab (III ECE)

EC320 VLSI Systems Lab (V ECE)

OFFICIAL RESPONSIBILITIES

Faculty Coordinator, Student Alumni Relationship Cell (SARC)

Member, Coordination Committee in Techno-Management Event "Techvibes"

Member, Training and Placement Cell

Member, Active Volunteer Committee, NSS

Member, Entrepreneurship Cell

Member, GP Birla Scholarship Cell

Member, IQAC Cell

Member, Coordination Committee in Techno-Management Event "Techvibes"

Coordinator, Internship Committee, IIC

## RESEARCH AREAS

VLSI Design, Nanotechnology, Analog Circuit Design, Technologies Beyond CMOS, CADENCE, Visual TCAD

## PUBLICATIONS

LIST OF PUBLICATIONS

BOOK CHAPTERS

1. Sharma, J., Sharma, P., Verma, D. (2025). Performance Comparison of 45 nm CMOS Based CDTA and 32 nm FinFET Based CDTA and Its Application as Universal Filter. In: Bhowmik, P.S., Bohre, A.K., Kirar, M.K., Kolhe, M.L., Suwarno, S. (eds) Application of Advance Techniques in Power and Energy Systems. CISTEE 2023. Lecture Notes in Electrical Engineering, vol 1305. Springer, Singapore. https://doi.org/10.1007/978-981-96-0476-0_13.

2. Sharma, P., Sharma, J., Sharma, R., Janyani, V., Verma, D. (2025). A Cervical Spine Motion Range Detection System Using MwCNTs-PDMS Based Sensor. In: Bhowmik, P.S., Bohre, A.K., Kirar, M.K., Kolhe, M.L., Suwarno, S. (eds) Application of Advance Techniques in Power and Energy Systems. CISTEE 2023. Lecture Notes in Electrical Engineering, vol 1305. Springer, Singapore. https://doi.org/10.1007/978-981-96-0476-0_22.

3. Sharma, J., Ahmad, R., Varma, T., Boolchandani, D. (2025). A 0.1–4.71 GHz Integer-N CP-PLL-Based Low-Power Frequency Synthesizer for High-Speed Applications. In: Gupta, A., Pandey, J.G., Chaturvedi, N., Dwivedi, D. (eds) Emerging VLSI Devices, Circuits and Architectures. VDAT 2023. Lecture Notes in Electrical Engineering, vol 1234. Springer, Singapore. https://doi.org/10.1007/978-981-97-5269-0_3.

4. Kumawat, M., Mathur, A., Sharma, J., Srinivasulu, A. "An Integrator Using Voltage Differencing Transconductance Amplifier," In Proceedings of 6th International Conference on Recent Trends in Computing. Lecture Notes in Networks and Systems, vol 177. Springer, (2021), Singapore. https://doi.org/10.1007/978-981-33-4501-0_1

5. Jyoti Sharma, Ritambhara and Avireni Srinivasulu, "A Novel CNFET Based CCCDTA and Its Application as a Schmitt Trigger" Advances in Smart Grid Automation and Industry 4.0, Lecture Notes in Electrical Engineering, vol 693. Springer, Singapore (2021), https://doi.org/10.1007/978-981-15-7675-1

6. Anushree Mathur, Mrinalini Kumawat, Jyoti Sharma and Avireni Srinivasulu, "A Schmitt Trigger by means of a Voltage Differencing Transconductance Amplifier" Advances in Smart Grid Automation and Industry 4.0, Lecture Notes in Electrical Engineering, vol 693. Springer, Singapore, (2021), https://doi.org/10.1007/978-981-15-7675-1

7. Jyoti Sharma, Mohd. Samar Ansari, and Sudip Kundu, "Tunable Universal Filter in ±0.5V 32nm CNFET for ISM 2.4 GHz Bluetooth/Zigbee Transceivers", Advances in Intelligent Systems and Computing book series of Springer, Vol. 624, 2018, ISBN: 978-981-10-5903-2_3 (Print) 978-981-10-1708-7 (Online), 10.1007/978-981-10-5903-2_3

8. Jyoti Sharma and Mohd. Samar Ansari, "Low THD ±0.75V 32nm CNFET Quadrature VCO for PLL and Costas-Loop Applications," Advances in Intelligent Systems and Computing book series of Springer, Vol. 479, 2017, ISBN: 978-981-10-1707-0 (Print) 978-981-10-1708-7 (Online), 10.1007/978-981-10-1708-7_54

SCI JOURNALS

1. Rajput, A., Kumawat, R., Srinivasulu, A., and Sharma, J. Robust and Compact Reversible Logic Gate for Low-Power and High-Performance Computing. Scientific Reports, (July 2025). DOI: 10.21203/rs.3.rs-7090231/v1 (Q1 SCIE, SCOPUS) Impact Factor: 3.9

2. Singhal, A., Sharma, J., Rajput, D.S. et al. Optimization of performance parameters of differential ring oscillator using Taguchi DoE and Pareto ANOVA techniques for fast-setting PLL frequency synthesizer. Analog Integr Circ Sig Process 124, 59 (2025). https://doi.org/10.1007/s10470-025-02468-z

3. Sharma, J., Singhal, A., Varma, T. et al. Design and Optimization of a Symmetrical Phase-Frequency Detector (SPFD) Through Statistical Techniques. Circuits Syst Signal Process (2025). https://doi.org/10.1007/s00034-025-03030-y, 11.02.2025

4. Jyoti Sharma, Riyaz Ahmad, Ashutosh Yadav, Tarun Varma and Dharmendar Boolchandani, "Design and Optimization of Phase Frequency Detector through Taguchi and ANOVA Statistical Techniques for Fast Settling Low Power Frequency Synthesizer," Integration, The VLSI Journal, Vol. 96, May 2024, 102162, https://doi.org/10.1016/j.vlsi.2024.102162

5. Jyoti Sharma, Gaurav Kumar Sharma, Tarun Varma and Dharmendar Boolchandani, "Optimization of performance parameters of phase frequency detector using Taguchi DoE and Pareto ANOVA techniques," Journal of Circuits, Systems, and Computers, Vol. 32, No. 09, (2023) 2350158, https://doi.org/10.1142/S021812662350158X

6. Jyoti Sharma, Gaurav Kumar Sharma, Tarun Varma and Dharmendar Boolchandani, "A high speed phase detection circuit with no dead zone suitable for minimal jitter and low power applications," Journal of Circuits, Systems, and Computers, Vol. 31, No. 15, (2022) 2250267, https://doi.org/10.1142/S021812662250267X

SCOPUS JOURNALS

1. Dua T., Kumawat, R., Singh N., Sharma, J., and Srinivasulu. "High Speed Energy Efficient Latch Architectures for Sequential Circuit Design", Journal of VLSI Circuits and Systems, 7(1), 22 March 2025, 19-24, ISSN: 2582-1458

2. Rajput, A., Kumawat, R., Sharma, J., and Srinivasulu, A. "Design of Novel High Speed Energy Efficient Robust 4:2 Compressor," Journal of VLSI Circuits and Systems, 6(2), 10 June 2024, 53–64, https://doi.org/10.31838/jvcs/06.02.06

3. Komal Shrivastavaa, Parul Garg, Avireni Srinivasulu and Jyoti Sharma, "A Sense Amplifier Based Modified Low Voltage Two Stage Flip-Flop with CNTFETs", ECS Transactions, Volume 107, Number 1, April 2022 (SCOPUS), 10.1149/10701.19517ecst, IF:0.7

4. Avireni Bhargava, J. K. Saini, Jyoti Sharma and K. M. Santhoshini, "A Review Article on Integrator Circuits Using Various Active Devices", Journal of Advance Research in Dynamical & Control Systems, Vol. 11, 26-32, 07-Special Issue, 2019 (SCOPUS).

5. Jyoti Sharma, Shantanu Chakraborty: MOCCCDTA-based Current Mode Tunable Universal Biquad Filter for Bluetooth Applications. International Journal of Electrical and Computer Engineering (IJECE), Vol 6, No 4: August 2016 (SCOPUS), http://doi.org/10.11591/ijece.v6i4.pp1499-1505

INTERNATIONAL JOURNALS

1. Jyoti Sharma, Saket Jhalani and Lakshya Dwivedi: Designing of Current-Mode Active Filter Using 45nm CMOS-Based CCII. IOSR Journal of VLSI and Signal Processing (IOSR-JVSP), Volume 6, Issue 2, pp 66-72, I (Mar-Apr. 2016).

2. Jyoti Sharma and Kshitij Sharma: Performance Evaluation of Electronically Tunable Filter Using CCCII in 45nm CMOS Technology for Wi-Fi Applications. International Journal of Science and Research (IJSR), Volume 5 Issue 3, 2053-2057, March 2016.

3. Jyoti Sharma and Anchal Sharma. Article: Universal Filter Design using 45nm CMOS-based DDCC for Bluetooth/ Zigbee Applications. International Journal of Computer Applications 134(13):36-40, January 2016. Published by Foundation of Computer Science (FCS), NY, USA.

INTERNATIONAL CONFERENCES

1. Jyoti Sharma, Archana Singhal, Tarun Varma and Dharmendar Boolchandani, "Design and Evaluation of a Power Efficient Phase-Frequency Detector with PVT Tolerance in 45 nm CMOS Technology", 3rd International Conference on Power Electronics, Intelligent Control, and Energy Systems (IEEE-ICPEICES-2024), DTU, Delhi, 26th -28th April, 2024.

2. A. Singhal, J. Sharma, D. S. Rajput, D. Boolchandani and C. Periasamy, "Design and Analysis of Fast Settling Wideband PLL Using a Novel Low Mismatch Current Charge Pump," 2024 IEEE Third International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES), Delhi, India, 2024, pp. 1045-1049, doi: 10.1109/ICPEICES62430.2024.10719110.

3. Jyoti Sharma and Puneet Sharma, "A comprehensive review of phase frequency detectors: Types, Characteristics and Applications", Recent Trends in Engineering and Sciences (RTES-2023), May 02-03, 2023, SVNIT Surat, India.

4. Jyoti Sharma, Tarun Varma and Dharmendra Boolchandani, "A brief review of the various phase-frequency detector architectures," 2021 IEEE International Symposium on Smart Electronic Systems (iSES), 2021, pp. 74-78, doi: 10.1109/iSES52644.2021.00028.

5. Apoorva Agarwal and Jyoti Sharma, "CMOS cum CNFET Based High Performance Differential Amplifier for Nanoelectronics", International Conference on Recent Innovations in Engineering and Technology, 18-19 February, 2017, MAIET, Jaipur.

6. Sharma, Jyoti; Ansari, Mohd. Samar; Sharma, Jankiballabh, "Electronically Tunable Resistor-less Universal Filter in ±0.5V 32nm CNFET," Electronic System Design (ISED), 2014 Fifth International Symposium on, vol., no., pp.206,207, 15-17 Dec. 2014 doi: 10.1109/ISED.2014.50 ISBN No. 978-1-4799-6964-7. Available on IEEE Xplore.

7. Sharma, J.; Ansari, M.S.; Sharma, J., "Current-Mode Electronically Tunable Resistor-less Universal Filter in ±0.5V 32Nm CNFET," Devices, Circuits and Communications (ICDCCom), 2014 International Conference

on, vol., no., pp.1,6, 12-13 Sept. 2014 doi: 10.1109/ICDCCom.2014.7024710 ISBN No. 978-1-4799-6052-1. Available on IEEE Xplore.

NATIONAL CONFERENCES

1. Saket Jhalani, Kshitij Sharma and Jyoti Sharma, "Comparison & Implementation of CCCII in 45nm CMOS for design of Oscillator" National Conference on New Vistas of Development in Communication Technology (NVDCT-2016), MAIET, Jaipur, 9th & 11th, April, 2016.

2. Shantanu Chakraborty and Jyoti Sharma, "Design of Universal filter using CDTA in 45nm CMOS Technology", National Conference on New Vistas of Development in Communication Technology (NVDCT-2016), MAIET, Jaipur, 9th & 11th, April, 2016.

3. Rohit Taxak and Jyoti Sharma, "Performance Evaluation of ICCII using CMOS and CNFET Technology", National Conference on Recent Technological Developments in Electronics and Electrical Engineering (RTDEEE-2016), JECRC, Jaipur, 12-13, August, 2016.

## INVITED TALKS

1. Delivered an invited talk on "Advances in VLSI Design: Techniques and Emerging Technologies" in the 'Demystifying VLSI Technology' lecture series, organized by the Department of Electronics and Communication, JECRC, Jaipur in collaboration with the IEEE Rajasthan Subsection and IEEE WIE Delhi Section on 07/04/025.

2. Hands-on training on LaTeX in 5-day online workshop on "Scientific Writing using LaTeX and Publication Ethics (SWLPE-2025)" organized at BIT Mesra, Jaipur Campus during 17th – 21st March 2025.

## WORKSHOPS ORGANISED

1. Convenor of 5 day online workshop on Scientific Writing using LaTeX and Publication Ethics organised at BIT Mesra, Jaipur Campus during 17th – 21st March 2025.

2. Coordinator in "Short Term Training Programme on VLSI Design using CADENCE Tool," 20/02/2024-21/03/2024 (one month).

## WORKSHOPS/ FDPs ATTENDED

1. One week online FDP on VLSI Testing and Testability, jointly organized by Electronics & ICT Academy, MNIT Jaipur from 14th July, 2025- 21st July, 2025.

2. One week (40-hour) online training program cum FDP on Foundations of VLSI Design Verification, jointly organized by Electronics & ICT Academy, MNIT Jaipur, IIITDM Jabalpur, IIT Roorkee and NIT Patna from 23th June, 2025- 5th July, 2025.

3. Two weeks online FDP on Semiconductor Devices Circuits & Sensor: Applications & Research Perspective, jointly organized by Electronics & ICT Academy, NIT Patna, IIITDM Jabalpur, MNIT Jaipur, IIT Kanpur and IIT Roorkee by the MeitY, Government of India from 9th June, 2025- 20th June, 2025.

4. One week UHV-II Course on Inculcating Universal Human Values in Technical Education, organized by AICTE, 25th to 30th December, 2024

5. Five days workshop on Emerging Cutting Edge Technologies in CMOS: Trends and Challenges, organized by NIT, Silchar, Assam, 11th to 15th December 2024.

6. One week UHV-I Course on Inculcating Universal Human Values in Technical Education, organized by AICTE, 4th to 8th December, 2023.

7. Two days FDP workshop on Recent Trends in MEMS for 6G Communications (RTMC-2023) organized by CSIR-CEERI, Jaipur Centre, August 17-18, 2023.

8. One-week workshop on VLSI to System Design: Silicon to End Application Approach organized by AICTE, Arm Education and STMicroelectronics, July 31st to 4th August, 2023.

9. One week outreach programme on Microelectronics and VLSI Design (MVD-2023) held during 15th - 19th May 2023, organized by Department of Electronics and Communication Engineering, BIT Mesra, Ranchi.

10. Two Week Online Faculty Development Program/Short term Training Program on Advanced Optimization Techniques and hands-on with MATLAB/SCILAB (AOT 2022), Supported by Ministry of Electronics and Information Technology (MeitY), Electronics & ICT Academies at MNIT Jaipur, NIT Patna & IIITDM Jabalpur, August 8 – 19, 2022.

11. One week online STTP on Emerging Tools and Techniques in VLSI, MEMS & MOEMS, Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur, 25/1/2022 to 29/1/2022.

12. Online training Programme on Digital Tools for Writing, Authoring and Reviewing Manuscripts, Supported by Ministry of Electronics and Information Technology (MeitY), Electronics & ICT Academies MNIT Jaipur, NIT Patna & IIT Guwahati, 21st September and 2nd October 2020.

13. Webinar on Issues in Nanoscale VLSI Circuit Design organized by Department of Electronics Engineering, University Departments, Rajasthan Technical University, Kota on 5 September 2020.

14. TEQIP-III sponsored webinar on Gender Equality and Women Rights, organized by RTU, Kota and Poornima College of Engineering, Jaipur on 17 July 2020.

15. Nine Weeks course on Introduction to Programming with MATLAB, Coursera, Vanderbilt University, June-July, 2020.

16. One week FDP on Tools and Techniques for Research in Engineering & Management, VGU, Jaipur, 23/06/2020-27/06/2020.

17. Webinar on Effective Research Manuscript Writing organized by Manipal University, Jaipur on 16-Jun-2020.

18. Webinar on Art and Challenges of Writing Papers for IEEE Transactions organized by IEEE Bangalore Section on 08 June, 2020.

19. Webinar on Recent Trends & Challenges in VLSI Design Techniques organized by Poornima College of Engineering, Jaipur on June 08, 2020.

20. One Professional Development Hour by attending Webinar on Endofullerenes: Nano-scale test tubes for single molecules and atoms organized by The National Academy of Sciences, India (NASI) - Delhi Chapter & MHRD-Institution Innovation Council (IIC) Deen Dayal Upadhyaya College Chapter on June 08, 2020.

21. Webinar on Education for Gen Z organized by Manipal University, Jaipur on 06-Jun-2020.

22. One Professional Development Hour by IEEE Electron Device Society (EDS) Distinguished Lecture (DL) on FOSS TCAD/EDA tools for Compact/SPICE Modeling on June 03, 2020.

23. One week Academy Training Programme on Latex and Technical Writing, MNIT, Jaipur, January 24-28, 2018.

24. Eight weeks course on Digital Signal Processing, NPTEL, IIT Kharagpur, July to September 2016.

25. One week Academy Training Programme on Digital CMOS Devices & Circuits, MNIT, Jaipur, 17/09/16-21/09/16.

26. One week Academy Training Programme on Digital Hardware Design & FPGA, MNIT, Jaipur, 3/09/16-16/09/16.

27. One week Academy Training Programme on VLSI Digital Signal Processing Systems, MNIT, Jaipur, 23/01/16-27/01/16.

28. One week STTP on Advanced Digital Devices and Systems, MNIT, Jaipur, 28/03/15-01/04/15.

29. Online course on Signals and Systems organised by IIT Bombay. Durations two Weeks: 12/03/15-13/05/15.

## MEMBER OF PROFESSIONAL BODIES

Senior Member, IEEE (Membership no. 95358399)

Lifetime Member of IETE

## TEXT AND REFERENCE BOOKS

1. Fundamentals of Communication, Genius Publications, 2008

2. Digital Communication (ECE) and Digital Communication and Information Theory (EEE), Genius Publications, 2008

3. Analog Communication, Genius Publications, 2008

4. Logic Synthesis and Electronic Design Automation, Ashirwad Publications, 2009

5. Computer Aided Design for VLSI, Ashirwad Publications, 2011

## INTELLECTUAL PROPERTY RIGHTS (IPR)

GRANTED

1. Puneet Sharma, Atul Kumar Sharma, Ritu Sharma, Vijay Janyani and Jyoti Sharma, "Method for Fabricating Flexible Strain-Sensor for Respiration Rate Detection and a Flexible Strain-Sensor Device Thereof," Application no. 202411056347 A, Issue no. 32/2024, 09/08/2024.

# Dr. Puneet Sharma

**Joined Institute on: 1-Feb-2010**

**Assistant Professor**

**Electronics and Communication Engg**

**Ph.D. MNIT Jaipur**

## CONTACT ADDRESS

405 ARRWA Jagatpura

Phone Office: 8852832501

Phone Residence: 8852832501

Email: puneetsharma@bitmesra.ac.in

## WORK EXPERIENCE

Teaching: 16 Years

## PROFESSIONAL BACKGROUND

Dr. Puneet Sharma is an Assistant Professor in Electronics and Communication Engineering at Birla Institute of Technology, Mesra – Jaipur Campus, where he has been serving since June 2017. He brings over fourteen years of academic, research, and administrative experience, with a strong focus on teaching, applied research, and technology development.

He received his B.E. in Electronics Engineering from the University of Rajasthan (2009), followed by an M.Tech. from MNIT Jaipur (2015). He earned his Ph.D. from MNIT Jaipur, with his doctoral research focused on "Synthesis and Characterisation of Polymer-Based Sensors for Electronic Applications." His research work emphasizes material innovation, sensor design, and real-world applicability.

Dr. Sharma's research contributions span flexible and wearable sensors, polymer nanocomposites, nanomaterial-based electrodes, biomedical instrumentation, and embedded sensor systems. He has actively worked on MWCNT-, graphene-, and PDMS-based flexible sensors, particularly for biomedical and healthcare monitoring applications. His work has resulted in multiple journal publications, conference papers, and granted patents, including a granted Indian patent (2025) on a flexible strain sensor for respiration rate detection, reflecting the translational and application-driven nature of his research.

In addition to academic research, Dr. Sharma has been deeply involved in industry-oriented and innovation-driven activities. He has coordinated student projects, internships, and startup initiatives, and has actively contributed to the development of incubation and entrepreneurship ecosystems at the institutional level. His engagement includes mentoring student startups, coordinating industry collaborations, and supporting technology commercialization efforts.

Dr. Sharma has also held several key administrative and leadership roles, including Technical Manager of an NABL-accredited calibration laboratory, Training and Internship Coordinator, Project Coordinator for undergraduate students, Startup Cell Coordinator, and mentor to multiple technical and innovation clubs. Through these roles, he has contributed to curriculum enrichment, laboratory development, quality assurance, and student skill development.

He actively participates in national and international research collaborations, workshops, and faculty development programs, and has delivered expert talks at reputed institutions. His current interests include expanding international collaborations, particularly in Indo-European research frameworks, and advancing industry-relevant, patent-driven, and socially impactful technologies.

## RESEARCH AREAS

FLEXIBLE AND WEARABLE SENSORS

Design, fabrication, and characterization of stretchable and skin-conformal sensors for physiological and biomedical monitoring applications.

POLYMER-BASED AND ORGANIC ELECTRONIC MATERIALS

Development of polymer and organic materials for electronic and sensing applications, with emphasis on flexibility, durability, and biocompatibility.

NANOMATERIAL-ENABLED SENSORS

Utilization of nanomaterials such as MWCNTs, graphene nanoplatelets (GNPs), and polymer nanocomposites to enhance sensitivity, stability, and performance of sensors.

BIOMEDICAL INSTRUMENTATION AND HEALTHCARE ELECTRONICS

Design of low-cost, reusable, and fast-response sensor systems for biomedical signal acquisition and healthcare monitoring.

FLEXIBLE ELECTRODES AND BIO-SIGNAL ACQUISITION

Development of novel flexible electrodes for ECG, respiration, and strain sensing applications.

EMBEDDED SYSTEMS AND SENSOR INTERFACING

Integration of sensors with microcontrollers and embedded platforms for real-time data acquisition, processing, and monitoring.

TRANSLATIONAL AND PATENT-ORIENTED RESEARCH

Focus on prototype development, intellectual property generation, and commercialization of sensor technologies.

## AWARD AND HONOURS

EXPERT SPEAKER INVITATIONS

Invited as an expert speaker to deliver technical and academic talks at reputed institutions, including MNIT Jaipur, for contributions in sensors, materials, and electronics research.

RESEARCH & INNOVATION RECOGNITION

Research work acknowledged through peer-reviewed journal publications, conference papers, and patents, highlighting impact in flexible sensors and biomedical instrumentation.

ACADEMIC AND ADMINISTRATIVE LEADERSHIP RECOGNITION

Recognized at the institutional level for contributions as Technical Manager (NABL Calibration Lab), Training & Internship Coordinator, Startup Cell Coordinator, and Project Coordinator, supporting academic excellence and innovation culture.

MENTORSHIP AND STARTUP SUPPORT RECOGNITION

Appreciation for mentoring student-led innovation and startup activities, including national-level participation and investor recognition through entrepreneurship platforms.

## PUBLICATIONS

JOURNAL PUBLICATIONS

1. Sharma, P., Sharma, R., Janyani, V., & Verma, D. (2023). Development of a multi-modal graphene nanoparticles (GNP)–polydimethylsiloxane (PDMS) flexible sensor for human activity monitoring and health assessment. International Journal of Electrochemical Science, 18(9), 100236.

2. Sharma, P., Baloda, S., Verma, D., Janyani, V., Sharma, R., & Gupta, N. (2024). Multiwall carbon nanotube/polydimethylsiloxane composites-based dry electrodes for bio-signal detection. IEEE Journal on Flexible Electronics, 3(3), 108–114.

3. Sharma, A. K., Sharma, A. K., Sharma, R., Sharma, P., & Singh, S. (2024). Multifunctional transparent conductive flexible sensor based on graphene/polyaniline/graphene sandwich composite on PDMS substrate. IEEE Sensors Journal.

4. Baloda, S., Sriram, S. K., Sharma, P., Singh, S., & Gupta, N. (2024). Development of SWCNTs/PDMS composite strain sensors integrated smart glove for human–machine interface applications. IEEE Sensors Journal.

5. Sharma, A. K., Sharma, A. K., Sharma, R., Sharma, P., & Sharma, M. D. (2025). Spray-coated GNP–PDMS flexible patch antenna-sensor for wireless wearable applications. IEEE Transactions on Materials for Electron Devices.

6. Sharma, P., Baloda, S., Verma, D., Janyani, V., Sharma, R., & Gupta, N. (2025). Flexible resistive strain sensor based on PDMS-coated MWCNTs and MWCNTs/GNP for wrist pulse and respiration monitoring. IEEE Journal on Flexible Electronics.

CONFERENCE PUBLICATIONS

1. Sharma, P., Sharma, R., Janyani, V., & Varshney, A. (2022). Design and simulation of PDMS-based flexible sensor using COMSOL and interfacing it for controlled robotics' applications. In Proceedings of the 6th IEEE International Conference on Condition Assessment Techniques in Electrical Systems.

2. Sharma, J., Sharma, P., & Verma, D. (2023). Performance comparison of 45 nm CMOS-based CDTA and 32 nm FinFET-based CDTA and its application as a universal filter. In Proceedings of the International Conference on Computational Intelligence and Smart Systems.

3. Sharma, P., Sharma, J., Sharma, R., Janyani, V., & Verma, D. (2023). A cervical spine motion range detection system using MWCNTs-PDMS-based sensor. In Proceedings of the International Conference on Computational Intelligence and Smart Systems.

4. Singh, B., Sharma, P., & Verma, D. (2023). Design and implementation of an IoT-enabled ECG sensor system for real-time cardiac monitoring. In Proceedings of the International Conference on Energy Systems, Drives and Automations (pp. 153–164).

## WORKSHOPS AND FDPs

1. Faculty Development Program through video conferencing, during 23-06-2020 to 27-06-2020.

2. Short Term Course on "Wireless Communication and Signal Processing Techniques" held during 20-05-2019 to 24-05-2019 at MNIT Jaipur, Sponsored by TEQIP-III.

3. FDP on "Advances in Optical Communication" held during 09-02-2019 to 14-02-2019 under the aegis of Electronic & ICT Academy, MNIT Jaipur. Endorsed by AICTE/UGC/NBA.

4. Short Term Course on "Scientific Techniques and Computational Approaches" held during 24-12-2018 to 28-12-2018 at MNIT Jaipur sponsor

5. FDP on "Computation Techniques on Optics and Photonics" held during 20-08-2018 to 25-08-2018 under the aegis of Electronic & ICT Academy, MNIT Jaipur. Endorsed by AICTE/UGC/NBA.

6. Short Term Course on "Communication Skills" during 14-05-2018 to 18-05-2018 at MNIT Jaipur by AICTE.

7. Workshop on "ADVANCES IN WIRELESS COMMUNICATION NETWORKS", during 29-09-2015 to 03-10-2015 in the Department of Electronics and Communication Engineering, by NIT, WARANGAL.

8. Workshop on "ENTREPRENEURSHIP DEVELOPMENT", during 15-01-2013 to 27-01-2013 in Birla Institute of Technology, Jaipur.

9. MHRD sponsored short term course on Basic Electronics, 28-06-2011 to 08-07-2011, by Indian Institute of Technology, Bombay.

10. QIP sponsored "SUMMER FACULTY RESEARCH FELLOW PROGRAMME", 13-05-2010 to 15-07-2010, by Indian Institute of Technical, Delhi.

## MEMBER OF PROFESSIONAL BODIES

MEMBERSHIPS

IEEE (Member, Sensor Council)

Member No.: 100011452

Grade: Member

Status: Active

Region: R10 - Asia and Pacific

Section: Rajasthan Subsection

IEEE Sensors Council – Active

Quantum Technical Community, IEEE Computer Society – Active

## CURRENT SPONSORED PROJECTS

Sponsored Project (DTST): "Design and Development of NIRBHAYA a Women Safety Device"

Status: Completed

## INTELLECTUAL PROPERTY RIGHTS (IPR)

1. GRANTED PATENT

Title: Method for Fabricating Flexible Strain-Sensor for Respiration Rate Detection and a Flexible Strain-Sensor Device Thereof

Inventors: Dr. Puneet Sharma, Dr. Atul Kumar Sharma, Dr. Ritu Sharma, Prof. Vijay Janyani, Jyoti Sharma

Application Number: 202411056347 A

Filing Date: 24 July 2024

Date of Publication: 09 August 2024

Date of Grant: 24 March 2025

Field of Innovation: Flexible Sensors / Wearable Sensors

Status: Granted

Description: This patent discloses a novel fabrication method for a flexible strain sensor intended for respiration rate detection and physiological monitoring. The invention focuses on a polymer-based flexible sensor architecture, enabling high sensitivity, mechanical flexibility, and stable performance under repeated strain conditions. The technology is suitable for wearable healthcare devices, biomedical monitoring systems, and real-time physiological sensing, with strong potential for translational and first-aid diagnostic applications.

2. GRANTED PATENT

Title: A PDMS-Based Graphene/PANI/Graphene (G–P–G) Sandwich Transparent Conducting Flexible Sensor Device and Its Fabrication Method Thereof

Inventors: Dr. Atul Kumar Sharma, Dr. Ritu Sharma, Dr. Anup Kumar Sharma, Dr. Puneet Sharma, Surbhi Singh

Patent Number: 568297

Application Number: 202411037200

Filing Date: 10 May 2024

Date of Grant: 02 July 2025

Field of Innovation: Flexible Sensors / Transparent Conducting Sensors

Status: Granted

Description: This patent presents a transparent and flexible sensor device based on a PDMS-supported graphene/PANI/graphene sandwich structure. The proposed architecture offers enhanced electrical conductivity, optical transparency, and mechanical flexibility, making it suitable for next-generation flexible electronics and wearable sensing platforms. The fabrication method is optimized for reliability, scalability, and compatibility with flexible substrates, supporting applications in smart wearables and flexible electronic systems.

# Snehanshu Shekhar

**Joined Institute on: 22-Sep-2011**

**Assistant Professor**

**Electronics and Communication Engg**

**Ph.D. (Pursuing-IIIT Kota), M.S (Research)-Microelectronics-TELECOM, Bretagne, France, M.E., B.E.**

## CONTACT ADDRESS

Jaipur

Phone Office: 0141-4059339

Phone Residence: 0141-4059339

Email: snehanshu.shekhar@bitmesra.ac.in

## WORK EXPERIENCE

Teaching: 15 Years

Research: 2.5 Years

Individual: 1 Years

## PROFESSIONAL BACKGROUND

1. Worked as Consultant at CEA, Grenoble, France

2. Worked as consultant at PSA Peugeot Citroen, Paris, France

3. Worked as consultant at ALTEN, Paris

## RESEARCH AREAS

Instrumentation and Control, NEMS, MEMS, Microprocessor and Microcontroller

## AWARD AND HONOURS

1. ALTEN & MEAE (France) Award

2. Fellowship for M.S. (Research)

3. P.G. Fellowship (MHRD, Govt. of India)

## PUBLICATIONS

INTERNATIONAL CONFERENCE/ SYMPOSIUM/ SEMINAR

1. Snehanshu Shekhar, "Lumped Control Virtual Instrumentation for Biomedical Applications", IJARSE, Volume 02, Issue 08, August 2013, pp-62-66.

2. Snehanshu Shekhar, Anchal Sharma, Shreyans Jain, Sahil Tangri, "A cost effective Communication device using morse code for disabled people", IJSRD, Volume 02, Issue 10, December 2014, pp-810-812.

3. Snehanshu Shekhar, Zairah Mustahsan, Gautam Saily, Navnika Wason, "Gesture recognition system based on 3-D Accelerometer", International Conference on Innovative Advancements in Engineering and Technology, organized by Jaipur National University in association with IETE, CEERI Pilani, DST Delhi, CSI, IEI (India), March 2014, pp-71.

4. Dheeraj Bhardwaj, Gitansh Gulati, Srishti Saraswat, Snehanshu Shekhar, Kanika Joshi and Komal Sharma, "Dual Band Multi Frequency Rectangular Patch Microstrip Antenna with Flyswatter Shaped Slot for Wireless Systems", American Institute of Physics (AIP): Scopus Indexed AIP Conference Proceedings 1715 (1), http://aip.scitation.org/doi/abs/10.1063/1.4942715.

5. Snehanshu Shekhar and Anupam Kumar, "Fractional Order Interval Type-2 Fuzzy Logic Controller", Recent Trends on Type-2 Fuzzy Logic Systems: Theory, Methodology and Applications; Springer, April 2023, Vol-425, pp 29-42; doi: https://link.springer.com/chapter/10.1007/978-3-031-26332-3_3

6. Snehanshu Shekhar, Anupam Kumar and Ashish Sharma, "A Systematic Literature Review on Mathematical Modelling and Verification of Human Cardiovascular System". In: Kumar Singh, K., Singh, S., Srivastava, S., Bajpai, M.K. (eds) Machine Vision and Augmented Intelligence. MAI 2023. Lecture Notes in Electrical Engineering, vol 1211. Springer, Singapore. https://doi.org/10.1007/978-981-97-4359-9_76

NATIONAL CONFERENCE/ SYMPOSIUM/ SEMINAR

1. Snehanshu Shekhar, Utkarsh Ranveer and Preeti Gupta, "Wireless Communication in Virtual Instrumentation for Biomedical Application", National Conference-Advanced Electronic & Wireless Communication Technologies organized by RCEW, Jaipur in association with ISTE & IETE, NCAEWCTB-005, March 2012, pp-08.

2. Dheeraj Bhardwaj, Deepak Verma, Snehanshu Shekhar and Komal Sharma "Design of Modified Broadband Microstrip Rectangular Shaped Slotted Antenna For WLAN Application" National Conference on Emerging Computing Technologies in modern ERA, held on S.S. Jain Subodh College on 22-23, January 2014.

3. Dheeraj Bhardwaj, Meenakshi Sharma, Snehanshu Shekhar and Komal Sharma "Broadband Triangular Patch Microstrip Antenna with Two Rectangular Slots for WLAN Application", National Conference on Emerging Computing Technologies in modern ERA, held on S.S. Jain Subodh College on 22-23, January 2014.

## MEMBER OF PROFESSIONAL BODIES

1. IEEE Senior Member

2. CSI Life Member

## CURRENT SPONSORED PROJECTS

1. Title of Project: Sudarshana

Description: Geofencing, GPS based AI supported women safety

Funding Amount: 50,000/-

Funding Agency: Birla Institute of Technology, Ranchi

## INTELLECTUAL PROPERTY RIGHTS (IPR)

1. Patent Title: An Efficient Power Distribution Based on IOT-Fog Resource for Effective Proxy Negotiation for Traffic Reduction

Patent No./Application No.: 2020104365

Award Date/Publication No. and Date: 03/03/2021

Agency/Country: IP Australia

Status: Granted

2. Patent Title: System on Chip Controller for battery charging for electric vehicle applications

Patent No./Application No.: 202211028719

Award Date/Publication No. and Date: (21/2022), 27/05/2022

Agency/Country: IP India

Status: Published

## FELLOW

Fellow-IETE

## MEMBER, EDITORIAL BOARD

EXTRAMURAL ASSOCIATION

1. Hony. Secretary, IETE Jaipur, Rajasthan Local Center (2024-26)

2. Invited Evaluator, School Innovation Contest, 2024, MIC, Govt. of India flagship program

3. SPOC- School Innovation Contest, 2023, MIC, Govt. of India flagship program

4. SPOC- Smart India Hackathon, 2023 & 2024, MIC, Govt. of India flagship program

5. SPOC- Dark Patterns Buster Hackathon (DPBH-2023), MIC, Govt. of India flagship program

6. Invited Scientific Jury member, NCSC 22 (held at Ahmedabad, Gujarat), a DST, Govt. of India, flagship program

7. Hony. Treasurer, IETE Jaipur, Rajasthan Local Center (2022-24)

8. Member, IETE Jaipur Center Executive Committee (2018-20) and (2020-22)

9. Member, Technical Program Committee, ICCT'17 (International Conference On Intelligent Communication And Computational Techniques)-IEEE Conference

10. Member, Technical Program Committee, ICCT'19 (International Conference On Intelligent Communication And Computational Techniques)-IEEE Conference

11. Reviewer, IEEE Computer Society, ICCT 2017. DOI:10.1109/INTELCCT.2017.832406

BIT ASSOCIATION

1. Member, Technical Program Committee, Techvibes (Technical fest of BIT Mesra, Jaipur Campus)-2016, 2017, 2018, 2023 & 2024

2. Coordinator, IETE Student Forum, BIT Mesra, Jaipur Campus

3. Member, Registration committee, BIT Mesra, Jaipur Campus

4. Coordinator, IIC Innovation Sub Committee, BIT Mesra, Jaipur Campus